Low-carbohydrate diets and all-cause and cause-specific mortality: Two cohort Studies

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Background Data on the long-term association between low-carbohydrate diets and mortality are sparse. Objective To examine the association of low-carbohydrate diets with mortality during 26 years of follow-up in women and 20 years in men. Design A prospective cohort study of women and men, followed from 1980 (women) or 1986 (men) until 2006. Low-carbohydrate diets, either animal-based (emphasizing animal sources of fat and protein), or vegetable-based (emphasizing vegetable sources of fat and protein) were computed from multiple validated food frequency questionnaire assessed during follow-up. Setting Nurses' Health Study and Health Professionals' Follow-up Study Participants 85,168 women (aged 34-59 years at baseline) and 44,548 men (aged 40-75 years at baseline) without heart disease, cancer, or diabetes. Measurement Investigator documented 12,555 deaths (2,458 cardiovascular, 5,780 cancer) in women and 8,678 deaths (2,746 cardiovascular, 2,960 cancer) in men. Results The overall low-carbohydrate score was associated with a modest increase in overall mortality in pooled analysis (Hazard Ratio, HR, comparing extreme deciles=1.12 (95% CI=1.01-1.24, p-trend=0.14). The animal low-carbohydrate score was associated with a higher all-cause mortality (pooled HR comparing extreme deciles=1.23, 95% CI=1.11-1.37, p-trend=0.05), cardiovascular mortality (corresponding HR=1.14, 95% CI=1.01-1.29, p-trend=0.029), and cancer mortality (corresponding HR=1.28, 95% CI 1.02-1.60, p for trend = 0.09). In contrast, a higher vegetable low-carbohydrate score was associated with lower all-cause (HR=0.80, 95% CI=0.75-0.85, p-trend<0.001) and cardiovascular mortality (HR=0.77, 95% CI=0.68-0.87, p-trend<0.001). Limitations Diet and lifestyle characteristics were assessed with some degree of error, however, sensitivity analyses indicated that results were not unlikely to be substantially affected by residual or confounding or an unmeasured confounder. In addition, participants were not a representative sample of the U.S. population. Conclusion A low-carbohydrate diet based on animal sources was associated with higher all-cause mortality in both men and women, whereas a vegetable-based low-carbohydrate diet was associated with lower all-cause and cardiovascular disease mortality rates. Primary funding source NIH grants CA87969, HL60712, and CA95589